that is, $\pi P = \pi$, where, for $\mu \in \mathcal{P}(\mathcal{X})$, we write μP for the probability measure

$$(\mu P)(A) := \int P(x, A)\, \mu(dx), \qquad A \in \mathcal{B}(\mathcal{X}). \tag{10}$$

The process $(X_t)_{t\geq 0}$ is called *stationary* if $X_0 \sim \pi$. We also use the notation $P^n(x, \cdot) := \delta_x P^n$ for $n \geq 0$. Recall that for a fixed $T > 0$,

$$\widehat{\mu}_T := \frac{1}{T} \sum_{t=0}^{T-1} \delta_{X_t} \tag{11}$$

denotes empirical *occupation measure*. For a measure μ and a measurable function $f : \mathcal{X} \to \mathbb{R}$, let $\mu(f) := \int f \, d\mu$.

Let $L^2(\pi)$ denote the Hilbert space of square-integrable functions $f : \mathcal{X} \to \mathbb{R}$, endowed with the inner product $\langle f, g\rangle_\pi := \pi(fg)$. Denote the subspace of π-centered functions by $L^2_0(\pi) := \{f \in L^2(\pi) : \pi(f) = 0\}$. For a separable Hilbert space $\mathcal{Y}$, we let $L^2(\mathcal{X}, \mathcal{B}(\mathcal{X}), \pi; \mathcal{Y})$, abbreviated $L^2(\pi; \mathcal{Y})$, denote the Bochner space of strongly $\mathcal{B}(\mathcal{X})$-measurable and Bochner square-integrable maps $f : \mathcal{X} \to \mathcal{Y}$ with

$$\|f\|^2_\pi := \|f\|^2_{L^2(\pi;\mathcal{Y})} = \int \|f(x)\|^2_{\mathcal{Y}}\, \pi(dx) < \infty. \tag{12}$$

Denote by $\mathcal{L}(\mathcal{H}_1, \mathcal{H}_2)$ be the space of bounded linear operators between two separable Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$, and $\mathcal{L}(\mathcal{H}_1)$ when $\mathcal{H}_1 = \mathcal{H}_2$. For $A \in \mathcal{L}(\mathcal{H}_1)$, denote the operator norm by $|\!|\!| A |\!|\!|_{\mathcal{H}_1} := \|A\|_{\mathcal{H}_1 \to \mathcal{H}_1}$. Let $\mathcal{L}_2(\mathcal{H}_1, \mathcal{H}_2)$ be the space of Hilbert–Schmidt operators, and $\mathcal{L}_2(\mathcal{H}_1)$ when $\mathcal{H}_1 = \mathcal{H}_2$, equipped with the norm $\|A\|^2_{\mathcal{L}_2(\mathcal{H}_1,\mathcal{H}_2)} = \sum_i \|Ae_i\|^2_{\mathcal{H}_2}$, where $\{e_i\}_i$ is any orthonormal basis of $\mathcal{H}_1$. We denote by I_d the $d \times d$ identity matrix.

Martingales. Let $(\mathcal{F}_t)_{t\geq 0}$ be a filtration. A sequence of $\mathcal{H}$-valued random variables $(d_t)_{t\geq 1}$ is called a *martingale difference sequence* with respect to $(\mathcal{F}_t)_t$ if d_t is $\mathcal{F}_t$-measurable and for every $t \geq 1$, $\mathbb{E}[d_t \mid \mathcal{F}_{t-1}] = 0$. Setting $M_0 := 0$ and $M_T := \sum_{t=1}^T d_t$ define an $\mathcal{H}$-valued *martingale*. Its *predictable quadratic variation* is

$$\langle M\rangle_T := \sum_{t=1}^T \mathbb{E}_{t-1}\|d_t\|^2, \tag{13}$$

where $\mathbb{E}_j[\cdot] := \mathbb{E}[\cdot \mid \mathcal{F}_j]$. For background on martingales and martingale difference sequences, we refer to Hall and Heyde (1980); Williams (1991).

Integral probability metrics. Let $\mathcal{F}$ be a class of measurable real-valued functions on $\mathcal{X}$. For $\mu, \nu \in \mathcal{P}(\mathcal{X})$ such that the integrals below are well defined, the *integral probability metric (IPM)* generated by $\mathcal{F}$ is

$$d_{\mathcal{F}}(\mu, \nu) := \sup_{f\in\mathcal{F}} |\mu(f) - \nu(f)|\,. \tag{14}$$

The choice of $\mathcal{F}$ determines the geometry on the space of probability measures. In particular, for the class

$$\mathcal{F}_{\mathrm{TV}} := \{f : \mathcal{X} \to [0, 1] : f \text{ is measurable}\}\,, \tag{15}$$

the corresponding IPM is the total variation distance:

$$d_{\mathcal{F}_{\mathrm{TV}}}(\mu, \nu) = \sup_{A\in\mathcal{B}(\mathcal{X})} |\mu(A) - \nu(A)| = \|\mu - \nu\|_{\mathrm{TV}}. \tag{16}$$

Suppose that $(\mathcal{X}, d)$ is a bounded space, i.e., $\mathrm{diam}(\mathcal{X}) < \infty$. Fix $x_0 \in \mathcal{X}$ and consider the anchored Lipschitz class

$$\mathcal{F}_{\mathrm{Lip}(d)} := \{f : \mathcal{X} \to \mathbb{R} : |f(x) - f(y)| \leq d(x, y)\ \forall x, y \in \mathcal{X},\ f(x_0) = 0\}\,. \tag{17}$$